# Spatiotemporal Interference of Cutaneous Shear Waves Dictates Tactile Perception in Ultrasonic Haptics

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**Abstract**

## 1 Introduction

Citation Test [1, 2]

## 2 Results

## 3 Discussion

## 4 Methods

## Acknowledgments

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## References

[1] Reardon, G., Dandu, B., Shao, Y. & Visell, Y. Shear shock waves mediate haptic holography via focused ultrasound. *Science Advances* (2023).

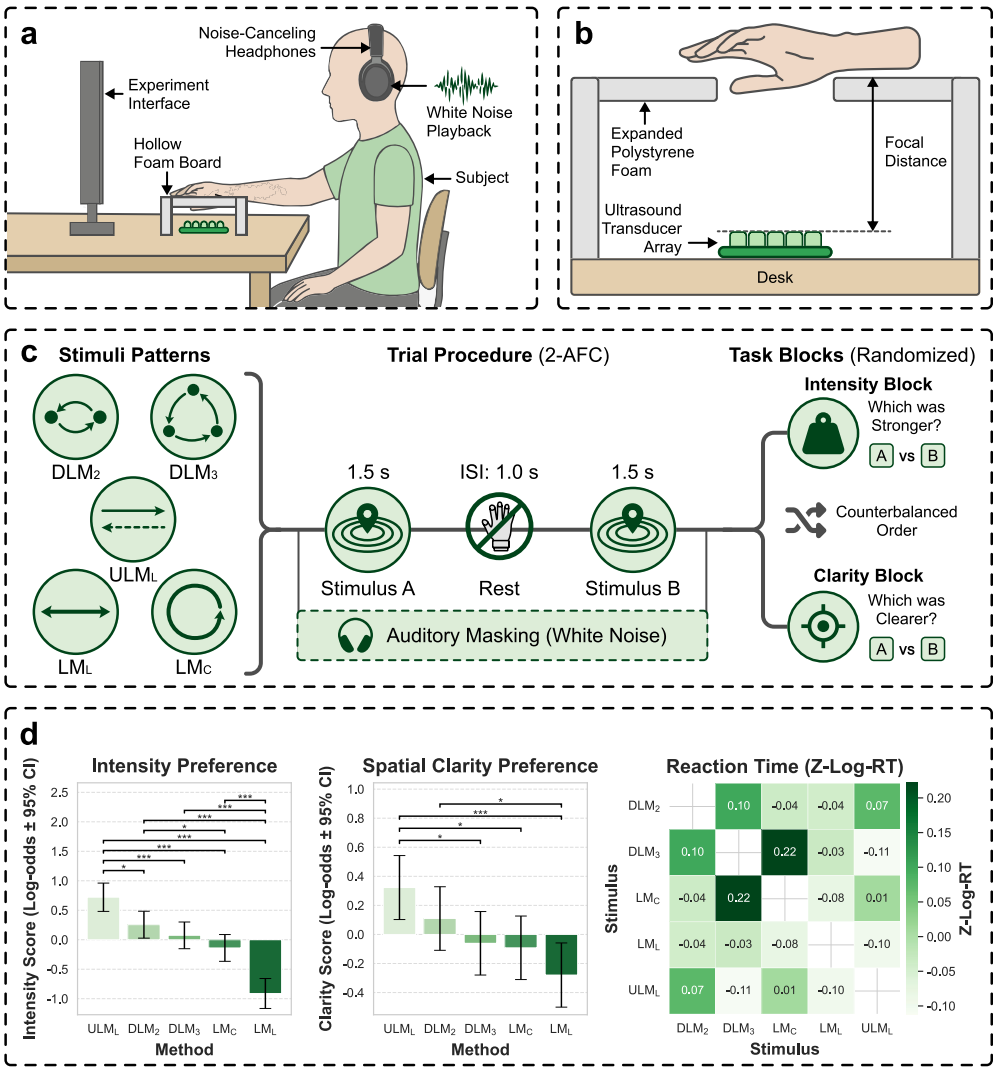


**Fig. 1**

[2] Takahashi, R., Hasegawa, K. & Shinoda, H. Tactile stimulation by repetitive lateral movement of midair ultrasound focus. *IEEE Transactions on Haptics* (2020).

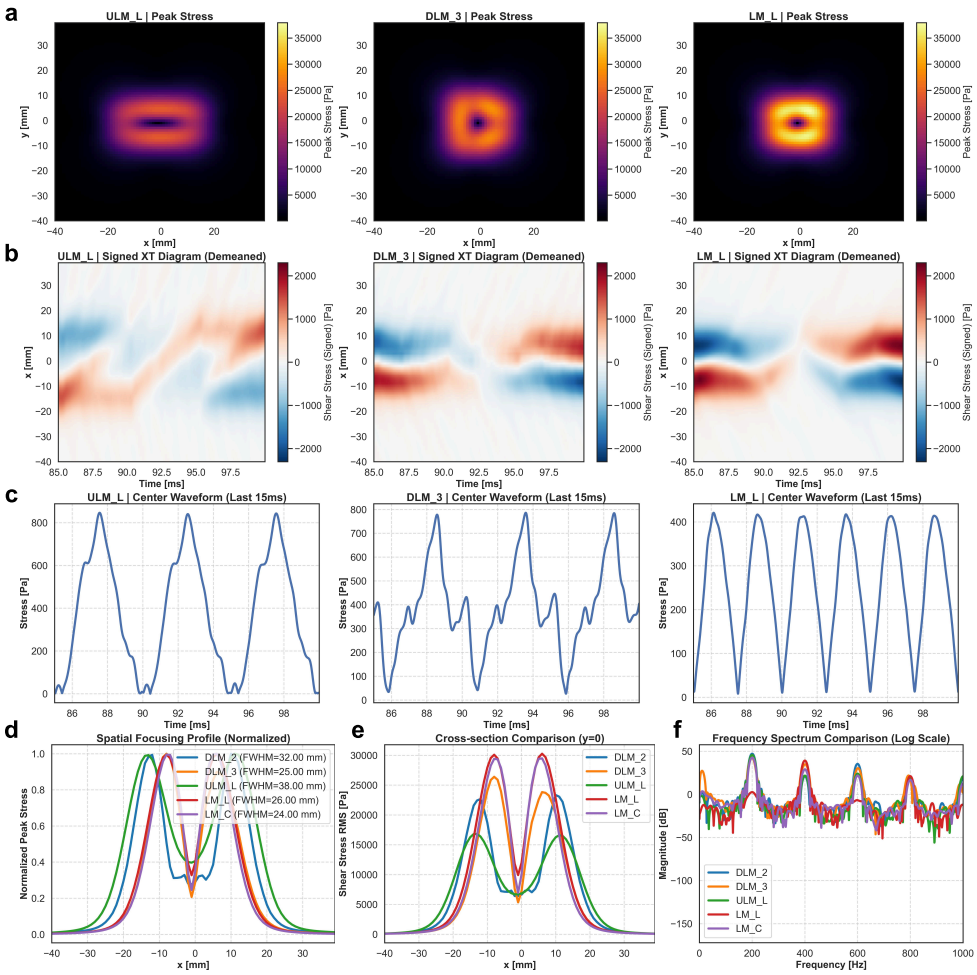
a
ULM_L | Peak Stress
DLM_3 | Peak Stress
LM_L | Peak Stress
y [mm]
x [mm]
Peak Stress [Pa]
b
ULM_L | Signed XT Diagram (Demeaned)
DLM_3 | Signed XT Diagram (Demeaned)
LM_L | Signed XT Diagram (Demeaned)
x [mm]
Time [ms]
Shear Stress (Signed) [Pa]
c
ULM_L | Center Waveform (Last 15ms)
DLM_3 | Center Waveform (Last 15ms)
LM_L | Center Waveform (Last 15ms)
Stress [Pa]
Time [ms]
d
Spatial Focusing Profile (Normalized)
DLM_2 (FWHM=32.00 mm)
DLM_3 (FWHM=25.00 mm)
ULM_L (FWHM=38.00 mm)
LM_L (FWHM=26.00 mm)
LM_C (FWHM=24.00 mm)
Normalized Peak Stress
x [mm]
e
Cross-section Comparison (y=0)
DLM_2
DLM_3
ULM_L
LM_L
LM_C
Shear Stress RMS [Pa]
x [mm]
f
Frequency Spectrum Comparison (Log Scale)
DLM_2
DLM_3
ULM_L
LM_L
LM_C
Magnitude [dB]
Frequency [Hz]

**Fig. 2**